Muhamad Akhdan Ramadhan

Computer Engineering Technology Student | Software, Networks & Embedded Systems

Bogor, Indonesia | m.akhdanramadhan@apps.ipb.ac.id
makhdanramadhan-ui.github.io | github.com/makhdanramadhan-ui

Profile

Fifth-semester Computer Engineering Technology student at Sekolah Vokasi IPB University. Combines hands-on software and AI automation practice with studies in networks, embedded hardware, industrial control, and data processing. Uses AI coding tools while directing implementation, auditing changes, and operating the systems. Interested in how these layers connect, from sensor input to application behavior.

Education

IPB University, Sekolah Vokasi | Bogor, Indonesia
Teknologi Rekayasa Komputer (Computer Engineering Technology), D4 applied bachelor's program
Semester 5 | Angkatan 61

Technical knowledge & practice

Software & web | Hands-on practice
Works with web interfaces, API integrations, persistent application state, and failure handling. Tools: JavaScript/Node.js, Python, HTML, CSS, SQL, SQLite, REST APIs, Git, Cloudflare Workers and D1.

AI systems & automation | Hands-on practice
Works with multi-agent orchestration, model routing, queued jobs, and audit flows that hold output when review is unavailable. Uses Linux services, monitoring, guarded fixes, and rollback to keep automation inspectable. Tools: LLM APIs, 9Router, systemd, Node.js, Python, Telegram controls, Docker Compose, n8n.

Computer networks | Coursework & practice
Studies routing, switching, and wireless fundamentals through Cisco coursework. Practices network access configuration and account-based user authentication with MikroTik. Tools and topics: Cisco networking, MikroTik, routing, switching, wireless access, authentication.

Embedded & hardware | Coursework & practice
Works with microcontrollers, sensor readings, and IoT communication, connecting physical inputs with software interfaces. Tools and topics: ESP/ESP32, Arduino, sensors, ESP-NOW, Firebase, device-to-dashboard data flow.

Industrial automation | Coursework
Studies PLC logic and the relationship between inputs, control conditions, and outputs. Uses CX-Programmer and Factory I/O to explore control sequences and simulated industrial processes.

Data & computing | Coursework
Learned data processing in RStudio and studied MATLAB, Linux, and computer architecture. Focuses on the relationship between data, the operating system, and the underlying computer.

Cisco Networking Academy course completion

CCNA: Switching, Routing, and Wireless Essentials | 17 Dec 2025
CCNA: Introduction to Networks | 29 Dec 2024
Course-completion certificates; not a professional CCNA certification.